

Reviewer Pack — Hypergraph Treewidth Inverse Proportional to DPLL Tree Size ...

Entry 923c457c4dd2 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Hypergraph Treewidth Inverse Proportional to DPLL Tree Size for Random 3-CNFs

Entry ID: 923c457c4dd2 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-09 16:43:04 UTC

1. Conjecture

Field A (mathematical branch): Hypergraph Theory **Field B** (complexity object): DPLL Search Tree

Statement:

For a random 3-CNF formula Φ with n variables, the treewidth of its hypergraph representation is inversely proportional to the DPLL tree size for Φ . Specifically, $E[\text{treewidth}(\Phi)] = \Theta(1 / E[\text{DPLL_tree_size}(\Phi)])$

Rationale (proposer's reasoning):

Hypergraph treewidth captures the structural complexity of the formula, which might influence the branching factor in DPLL search. Lower treewidth suggests more structured clauses, leading to smaller search trees.

Taxonomy category: PROOF_COMPLEXITY_EXT_FREGE (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): eb8687a79dd70a6d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.95	SAFE	SAFE

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.1s

5.1 Generated Python source

```
import random
from itertools import combinations

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3cnf(n):
        clauses = set()
        for _ in range(10 * n): # Generate enough clauses to ensure a dense formula
            variables = random.sample(range(1, n + 1), 3)
            clause = tuple(sorted(variables))
            if clause not in clauses:
                clauses.add(clause)
        return clauses

    def hypergraph_treewidth(clauses):
        # Simplified treewidth algorithm for demonstration purposes
        nodes = set()
        edges = []
        for clause in clauses:
            nodes.update(clause)
            for i in range(len(clause)):
                for j in range(i + 1, len(clause)):
                    edges.append((clause[i], clause[j]))

        def dfs(node, visited):
            if node in visited:
                return
            visited.add(node)
            for neighbor in edges[node]:
                dfs(neighbor, visited)

        max_clique_size = 0
        for subset in combinations(nodes, len(nodes) // 2 + 1):
            clique = set(subset)
            if all((node in clique or node not in nodes) for node in nodes):
                max_clique_size = max(max_clique_size, len(clique))

        return max_clique_size
```

```

def dpll_tree_size(clauses, assignment=None):
    if assignment is None:
        assignment = {}

    def solve():
        if all(any(lit in assignment and assignment[lit] == val for lit in clause) for clause in clauses):
            return 1
        if any(all(lit not in assignment or assignment[lit] != val for lit in clause) for clause in clauses):
            return 0

        var = next(var for var in range(1, n + 1) if var not in assignment)
        count_true = solve()
        if count_true > 0:
            return count_true
        assignment[var] = False
        count_false = solve()
        del assignment[var]
        return count_false

    return solve()

n = random.randint(5, 40)
clauses = generate_3cnf(n)
treewidth = hypergraph_treewidth(clauses)
dpll_size = dpll_tree_size(clauses)

if treewidth == 0 or dpll_size == 0:
    return {
        "metric_name": "treewidth * DPLL tree size",
        "metric_value": None,
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

metric_value = treewidth * dpll_size
return {
    "metric_name": "treewidth * DPLL tree size",
    "metric_value": metric_value,
    "instances_tested": 1,
    "conjecture_holds": True,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all(r["conjecture_holds"] for r in results):
        mean_value = sum(r["metric_value"] for r in results) / len(results)

```

```

std_value = (sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results)) *
support_fraction = 1.0
print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
else:
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conclusion"])
    counterexample = "mapping_undefined"
    print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test timed out before producing data; support fraction cannot be evaluated | next: Run test with extended timeout and higher resource allocation

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	110754
2	preregistration	ollama_remote	qwen3:8b	0	0	24165
3	novelty	ollama_remote	qwen3:8b	0	0	20694
4	novelty	ollama_remote	qwen3:8b	0	0	12506
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9798
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10257
7	judge	ollama_remote	qwen3:8b	0	0	15308

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 203481 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in [research/audit/923c457c4dd2.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice

3. The full LLM transcripts are in `research/audit/923c457c4dd2.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/923c457c4dd2.tar.gz` (if generated)